TabulaCortex: Towards Data Analysis in Spreadsheets

Sau Lai YIP, Shiwen WU

March 11th

香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

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The Spreadsheet Data Challenge

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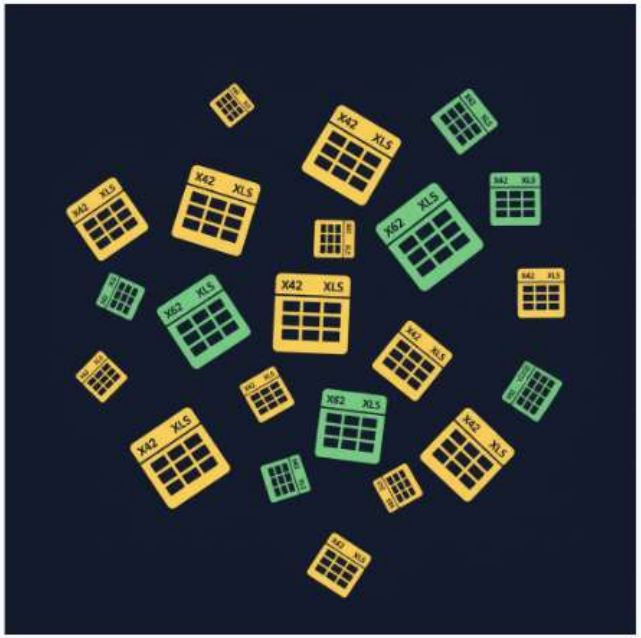

Fragmented

Hundreds of local files

Spreadsheet are widely used ...

Spreadsheets are the backbone of data management across organizations—flexible, familiar, and indispensable for tracking everything from budgets to inventories.

Fragmented Repositories

As organizations scale, data becomes scattered across hundreds of locally managed Excel files with no central coordination.

Inconsistent Structures

Varying formats, column names, and data conventions make cross-file analysis nearly impossible without manual intervention.

labor-intensive

Consolidating data for nationwide reporting becomes a labor-intensive process riddled with errors and delays.

System Overview

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Excel Files

→

Phase 1

Interactive DB Construction

1. AI agent analyzes uploaded files
2. Converses with user to resolve inconsistencies
3. Collaboratively designs unified relational schema

→

Unified Database

→

Phase 2

Conversational QA with NL2SQL

1. Users ask questions in natural language
2. Agent locates relevant tables and runs SQL
3. Returns analysis results

1. Interactive DB Construction

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1. AI-Driven Initial Design

Input: Raw Excel files
Action: Extracts metadata; builds base schema
Output: Initial schema design with merging suggestions

→

2. User-Driven Update

Agent Framework: **Tool calling agent**
Core Tools: Delete/Create/Merge
Core targets: Tables, Columns, FKs, Constraints

→

3. Database Creation

Output: SQLite database file

Common Updates

- Rename DB, Tables, Columns
- Modify constraints & datatypes
- Merge tables (with or without source tracking)
- Merge columns

Available Tools

```
delete_table_tool(cur_schema, table_identifier)
create_table_tool(cur_schema, table_entry)
merge_table_tool(cur_schema, table_identifier1, table_identifier2, table_identifier_new)
delete_column_tool(cur_schema, table_identifier, column_identifier)
create_column_tool(cur_schema, table_identifier, column_entry)
merge_column_tool(cur_schema, table_identifier, column_identifier1, column_identifier2,
column_identifier_new, transform_for_DB1="ORIGINAL", transform_for_DB2="ORIGINAL")
add_foreign_keys_tool(cur_schema, new_foreign_keys)
update_whole_foreign_keys_tool(cur_schema, new_whole_foreign_keys)
update_db_name_tool(cur_schema, new_db_name)
update_column_tool(cur_schema, table_identifier, column_identifier, field_name, new_value)
update_table_name_tool(cur_schema, table_identifier, new_table_name)
```

2. Conversational QA (NL2SQL)

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1. User Question

User selects a constructed DB and asks a question in natural language

2. Agent Processing

Tool calling agent framework processes the query

Available Tools

```
exec_sql_tool(sql_query)
read_nofordb_files_tool(file_names)
prepare_answer_tool()
```

Thanks!

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Demo 1

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TabulaCortex: Conversational Excel Analysis Platform

Transform your Excel files into queryable databases through conversational interaction.

DB Construction | Database Visualization | Analysis with QA

File Upload

Upload Excel/CSV Files

010营口市老边区"低保、低... .xlsx 22.2 KB

Progress Logs

Processing Progress

```
Found 2 sheet groups
Enriched sheet names
Processing 2 groups concurrently...
Processing group: individual_1
Processing group: individual_2
Completed schema design for group: individual_1
Completed schema design for group: individual_2
Designed 0 foreign key relationships
Database name: 营口市老边区低保低收入失能半失能老人护理补贴申请拨付
```

Clear Logs | Refresh

File Structure Analysis

010营口市老边区低保低收入失能半失能护理补贴申请拨付表_1.xlsx

- 低保: 低保失能、半失能老人护理补贴申请拨付汇总表 (rows: [4...103] | type: 1D_2level)

```
<COLUMNS>
- 区属
- 失能:
  - 人数
  - 标准
  - 金额
- 半失能:
  - 人数
  - 标准
  - 金额
- 小计:
  - 补贴金额总计
  - 市级匹配金额
```

Diagram Editor & Preview

Tips: Edit the code in the left panel and click "Update Changes" below. Can copy code to Mermaid Live for advanced editing.

Database Schema Design

低保失能半失能老人护理补贴申请拨付汇总表	
TEXT	区属
INTEGER	失能_人数
REAL	失能_标准
REAL	失能_金额
INTEGER	半失能_人数
REAL	半失能_标准
REAL	半失能_金额
REAL	小计_补贴金额总计
REAL	小计_市级匹配金额
REAL	小计_区级匹配金额

低收入失能半失能老人护理补贴申请拨付汇总表	
TEXT	区属
INTEGER	失能_人数
REAL	失能_标准
REAL	失能_金额
INTEGER	半失能_人数
REAL	半失能_标准
REAL	半失能_金额
REAL	小计_补贴金额总计
REAL	小计_市级匹配金额
REAL	小计_区级匹配金额

Mermaid Code Editor

Mermaid Code

```
erDiagram
    低保失能半失能老人护理补贴申请拨付汇总表 {
        TEXT 区属
        INTEGER 失能_人数
        REAL 失能_标准
        REAL 失能_金额
        INTEGER 半失能_人数
        REAL 半失能_标准
        REAL 半失能_金额
        REAL 小计_补贴金额总计
        REAL 小计_市级匹配金额
        REAL 小计_区级匹配金额
    }
    低收入失能半失能老人护理补贴申请拨付汇总表 {
```

Schema-File Mapping

This panel shows the mapping between designed schema and source files if there has been merging. You can edit it and click 'Update Changes' to apply.

Demo 1

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Conversation

DB construction Conversation

Optimization Opportunities

发现1个明确的表合并机会：两个结构完全一致的补贴申请表仅存在受益人类型差异，建议通过垂直合并+分类标识的方式整合数据。当前无其他结构优化需求。

Table Merge Recommendations:

1. Merge Tables: **低保失能、半失能老人护理补贴申请拨付汇总表, 低收入失能、半失能老人护理补贴申请拨付汇总表****

- **Affected sheets:** 010营口市老边区低保低收入失能半失能护理补贴申请拨付表_1.xlsx::低保, 010营市老边区低保低收入失能半失能护理补贴申请拨付表_1.xlsx::低收入
- **Strategy:** 建议将低保和低收入两类护理补贴申请数据合并为统一表结构，具体操作包括：
- 删除重复的表结构
- 创建统一表名如『护理补贴申请拨付汇总表』
- 增加『受益人类型』字段（低保/低收入）作为分类标识
- 保留原有4个核心字段（申请人信息、失能等级、补贴金额、拨付状态等）
- 通过新增分类字段实现数据隔离与查询

What's Next?

You can now:

- **Update the schema design**: Edit the schema using the Mermaid editor above or describe changes in the message box.

Description or Messages

please merge based on your suggestion

Start Initial Design | Update Changes | Undo Last Change | Create Database | Clear Chat

Demo 1

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DB construction Conversation

✅ Schema Update Complete!

📝 **Actions Performed:**

Action 1: merge tables with distinguishing column - Affected: 低保失能半失能老人护理补贴申请拨付汇总表; 低收入失能半失能老人护理补贴申请拨付汇总表

- Step 1: Merged tables 低保失能半失能老人护理补贴申请拨付汇总表 and 低收入失能半失能老人护理补贴申请拨付汇总表 into 护理补贴申请拨付汇总表
- Step 2: Created column: 受益人类型 in table 护理补贴申请拨付汇总表

Action 2: delete table - Affected: 低保失能半失能老人护理补贴申请拨付汇总表; 低收入失能半失能老人护理补贴申请拨付汇总表

- Step 1: Deleted table: 低保失能半失能老人护理补贴申请拨付汇总表
- Step 2: Deleted table: 低收入失能半失能老人护理补贴申请拨付汇总表

🗂️ **Updated Schema Summary:**

- **Database:** 营口市老边区低保低收入失能半失能老人护理补贴申请拨付
- **Tables:** 1
- **Foreign Keys:** 0

Tables:

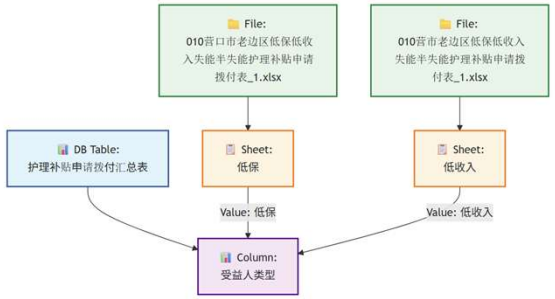
- 护理补贴申请拨付汇总表 (11 columns)

Description or Messages

Database Schema Design

护理补贴申请拨付汇总表		
TEXT	区属	
INTEGER	失能_人数	
REAL	失能_标准	
REAL	失能_金额	
INTEGER	半失能_人数	
REAL	半失能_标准	
REAL	半失能_金额	
REAL	小计_补贴金额总计	
REAL	小计_市级匹配金额	
REAL	小计_区级匹配金额	
TEXT	受益人类型	NOT NULL

File to Database Mapping (with merging)



Demo 1

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NL2SQL Agent Conversation

低收入群体中,失能人数和半失能人数的总数分别为多少？

Execution History

Step 1: SQL Execution

```
SELECT SUM(失能_人数) AS 总失能人数, SUM(半失能_人数) AS 总半失能人数 FROM 护理补贴申请拨付汇总表 WHERE 补贴类型 = '低收入';
```

Result:

```
总失能人数,总半失能人数
4930,4837
```

Final Answer

根据数据库中的执行结果，在低收入群体中：

- **失能人数**的总数为 **4,930人**
- **半失能人数**的总数为 **4,837人**

这些数据通过汇总护理补贴申请表中补贴类型为“低收入”的记录得出，计算方式为直接统计失能和半失能两类人群的数量总和。

Database Selection

Select Database
Choose from available databases in ./created_DBs/

营口市老边区低保低收入失能护理补贴

Database '营口市老边区低保低收入失能护理补贴' selected successfully

Database Path: created_DBs/营口市老边区低保低收入失能护理补贴/营口市老边区低保低收入失能护理补贴.sqlite

Refresh Database List

NL2SQL Agent Conversation

那低保

Execution History

Step 1: SQL Execution

```
SELECT SUM(失能_人数) AS 总失能人数, SUM(半失能_人数) AS 总半失能人数 FROM 护理补贴申请拨付汇总表 WHERE 补贴类型 = '低保';
```

Result:

```
总失能人数,总半失能人数
4431,4913
```

Final Answer

根据数据库中针对**低保群体**（补贴类型为“低保”）的统计结果：

- **失能人数**的总数为 **4,431人**
- **半失能人数**的总数为 **4,913人**

这些数据来源于护理补贴申请拨付汇总表中，筛选条件为补贴类型为“低保”的记录。需要注意的是，数据库中使用的是“低保”这一表述，可能是“低保”（最低生活保障）的另一种表述方式或简写。若存在其他分类需求，建议进一步确认补贴类型的定义或补充筛选条件。

Demo 2

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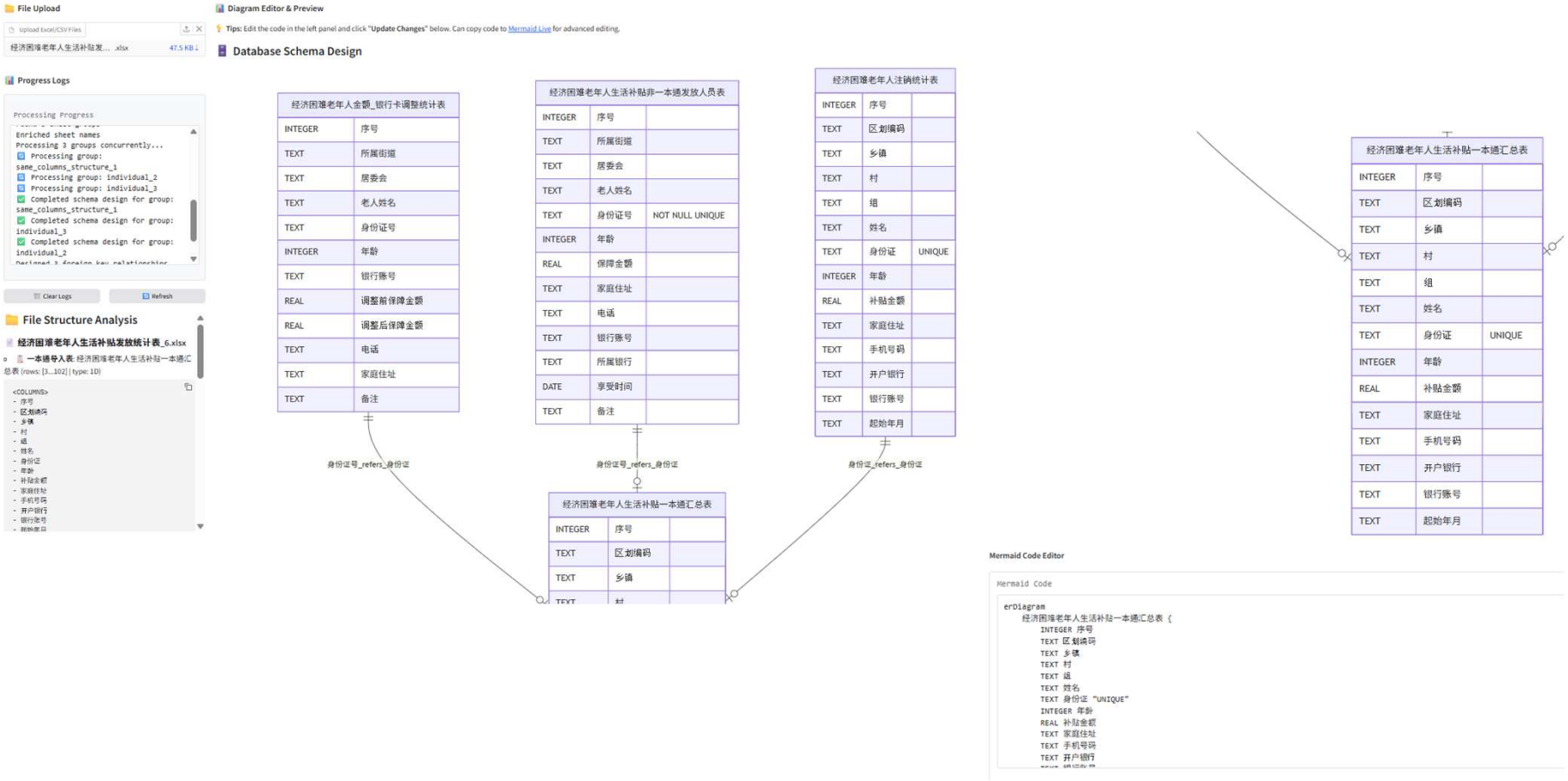

Demo 2

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Mermaid Code Editor

Mermaid Code

```
        TEXT 老人姓名
        TEXT 身份证号 "NOT NULL UNIQUE"
        INTEGER 年龄
        REAL 保障金额
        TEXT 家庭住址
        TEXT 电话
        TEXT 银行账号
        TEXT 所属银行
        DATE 享受时间
        TEXT 备注
    }
    经济困难老年人金额_银行卡调整统计表 ||--o| 经济困难老年人生活补贴一本通汇总表 : 身份证号_refers_身份证
    经济困难老年人生活补贴非一本通发放人员表 ||--o| 经济困难老年人生活补贴一本通汇总表 : 身份证号_refers_身份证
    经济困难老年人注销统计表 ||--o| 经济困难老年人生活补贴一本通汇总表 : 身份证_refers_身份证
```

delete

Demo 2

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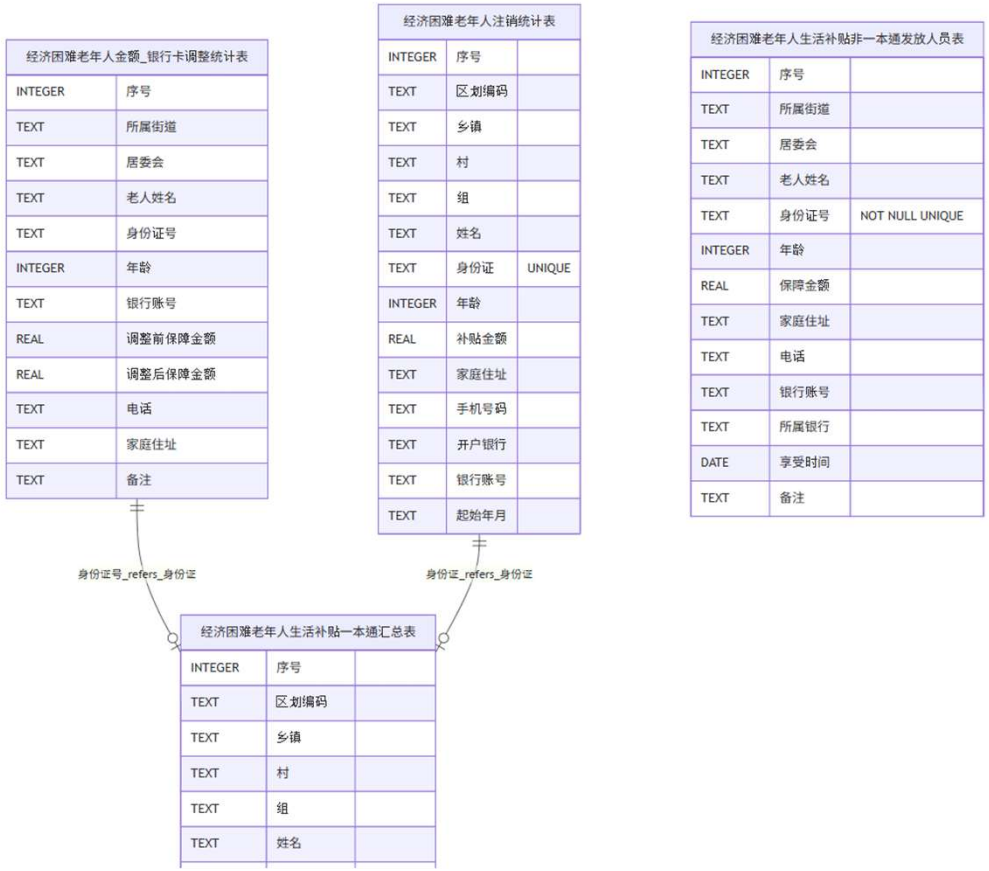

Schema Update Complete!

Actions Performed:

Action 1: update foreign_key - Affected: 经济困难老年人生活补贴非一本通发放人员表.身份证号; 经济困难老年人注销统计表.身份证

- Step 1: Updated foreign keys to 2 relationship(s)

Updated Schema Summary:

- **Database:** 经济困难老年人生活补贴发放统计系统
- **Tables:** 4
- **Foreign Keys:** 2

Tables:

- 经济困难老年人生活补贴一本通汇总表 (14 columns)
- 经济困难老年人注销统计表 (14 columns)
- 经济困难老年人金额_银行卡调整统计表 (12 columns)
- 经济困难老年人生活补贴非一本通发放人员表 (13 columns)

Demo 2

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Database Selection

Select Database
Choose from available databases in ./created_DBs/

经济困难老年人生活补贴发放统计

Database '经济困难老年人生活补贴发放统计' selected successfully

Database Path: created_DBs/经济困难老年人生活补贴发放统计/经济困难老年人生活补贴发放统计.sqlite

Refresh Database List

ent Conversation

保障金额发生调整的经济困难老年人中，调整后金额比调整前高的人数为多少？调整后金额比调整前低的人数为多少？

Execution History

Step 1: SQL Execution

```
SELECT
    SUM(CASE WHEN "调整后保障金额_银行卡号" > "调整前保障金额_银行卡号" THEN 1 ELSE 0 END) AS 增加人数,
    SUM(CASE WHEN "调整后保障金额_银行卡号" < "调整前保障金额_银行卡号" THEN 1 ELSE 0 END) AS 减少人数
FROM "经济困难老年人金额_银行卡调整统计表";
```

Result:

```
增加人数,减少人数
57,41
```

Final Answer

根据数据库中的统计结果，在保障金额发生调整的经济困难老年人中：

- **调整后金额比调整前高的老年人人数为57人**
- **调整后金额比调整前低的老年人人数为41人**

这一结果通过直接比较每位老年人调整前后的保障金额（银行卡账户数据）得出，统计逻辑由SQL查询验证，数据来源明确且计算过程无矛盾。若需进一步分析调整